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Case Study

A Novel Approach to Scavenging Anesthetic Gases in Rodent Surgery

Laboratory animal procedures using gas anesthetics may amass elevated waste gas concentrations in operating rooms if controls are not implemented for capturing and removing the vapors. Area sampling using an infrared analyzer indicated isoflurane concentrations likely to exceed occupational exposure guidelines. Our study showed environmental concentrations of oxygen as high as 40% and isoflurane concentrations >100 ppm when no controls or merely passive controls were utilized. These extraneous isoflurane emissions were determined to be originating from the pre-procedural induction process as well as the gas delivery nose cone. A novel waste gas collection cylinder was designed to enclose the gas delivery nose cone and animal head during the administration of anesthetic gases. The vented cylinder utilized a house vacuum to remove the waste anesthetic gases from the surgical field. A commercially available induction chamber designed to be actively and externally exhausted was used to lower concentrations during the induction process. With implementation of local exhaust ventilation controls, waste anesthetic gas concentrations decreased to below recommended occupational exposure levels. In vitro (sham) testing compared favorably to in vivo measurements validating the reduction capability of active ventilation during rodent anesthetic administration. In vivo isoflurane reductions for the induction chamber emissions, the operating room, and the surgeon's breathing zone were 95%, 60%, and 53%, respectively. The same measurements for an in vitro procedure were 98%, 84%, and 87%, respectively.

INTRODUCTION

Scavenging of waste gases is essential to protect laboratory personnel working with animals and volatile anesthetics.⁽¹⁾ The use of isoflurane anesthesia is increasing in animal research due to its short duration of action, safety to the animal, and the control of anesthetic depth. There is no specific occupational exposure limit established for isoflurane; however, the National Institute for Occupational Safety and Health (NIOSH) recommends that no worker should be exposed above concentrations greater than 2 parts per million (ppm) of any halogenated anesthetic agent over a sampling period not to exceed one hour.⁽²⁾ This occupational exposure limit (OEL) has great limitations because of differences in structures and metabolites between the compounds. The limitations have been noted in the literature and NIOSH responded with a request for information published in the *Federal Register*.^(3,4)

NIOSH intends to review and evaluate toxicity data for the halogenated anesthetic agents of isoflurane, desflurane, and sevoflurane.⁽⁵⁾ In 1977, NIOSH published a recommended exposure limit (REL) of 2 ppm as a 60-minute ceiling for halogenated gases. The halogenated anesthetic agents, isoflurane, desflurane, and sevoflurane were

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not included in this recommendation. These three halogenated anesthetics are now commonly used for anesthesia in modern hospitals; however, no OELs exist for them. As part of their review, NIOSH is requesting: (1) "comments and information relevant to the evaluation of health risks associated with occupational exposure to isoflurane, desflurane, and sevoflurane, (2) reports or other data that demonstrate adverse health effects in workers exposed to isoflurane, desflurane, and sevoflurane, and (3) information pertinent to establishing a REL for isoflurane, desflurane, and sevoflurane."⁽⁵⁾ Although there are no federal criteria, the State of California Code of Regulations, Title 8, Section 5155 establishes an isoflurane permissible exposure limit (PEL) of 2 ppm as an eight-hour time-weighted average (TWA).⁽⁶⁾

As with most solvents, an acute effect of isoflurane exposure is central nervous system depression. These acute symptoms include headache, nausea, fatigue, drowsiness, and irritability. Halogenated aliphatic hydrocarbons, such as isoflurane, are activated by a cytochrome-P450 enzyme to trifluoroacetylhalide which binds to protein and elicits an immune response in the body. This mechanism is believed to contribute to isoflurane-related chronic diseases such as hepatotoxicity. The enzyme is also involved with oxidizing ethanol and many other solvents and is expressed in the kidney, lung, lymphocytes, and bone marrow, as well as the liver, making all of these potential target organs. Isoflurane may cause cardiac output to be depressed by as much as 50%, producing arrhythmia leading to an increased risk of heart disease.⁽⁷⁾ Researchers have also reported causal relationships between waste anesthetic gas exposure and neurological and reproductive disorders.^(8,9)

The authors investigated an incident at a large biomedical research facility in which a researcher brought an active electrocautery tip near the breathing zone of a rodent undergoing anesthesia. The mixture used to anesthetize the rodent contained 100% compressed oxygen delivered through a commercially available rodent anesthesia system capable of delivering 0–5% of isoflurane via a precision vaporizer (EZ-Anesthesia, Euthanex Corporation, Palmer, Pa.). The result was ignition of the rodent and the plastic nose cone delivering the gases. This resulted in injury to the animal and a decision to euthanize. The technicians performing the procedure were not injured although they were distressed by the incident.

The incident investigation identified a number of hazards including oxygen levels well above normal ambient concentrations of 20.9 percent. Oxygen levels were measured as high as 40% near the head of the rodent and 33% at the incision site. Also, the investigation found that isoflurane concentrations exceeded 100 ppm (the upper limit of detection) at both the incision site and near the rodent head area. The authors used this information to develop procedures and controls to lower the oxygen level and flammability risk, and to decrease isoflurane exposures of employees.

Similar incidents and control alternatives have been reported in the literature.^(10,11) The focus of this article is to

present a novel approach for controlling waste anesthetic gases (WAG) in the rodent surgical setting. The authors have since identified similar hazards in rodent imaging and other non-surgical rodent procedures involving administration of anesthetic gasses.

MATERIALS AND METHODS

This study was requested by the Institutional Animal Use and Care Committee (IACUC) and was conducted in accordance with federal animal care guidelines in an Association for Assessment and Accreditation of Laboratory Animal Care International (AAALAC)-accredited institution. These investigations were conducted opportunistically with animals that were undergoing anesthesia as part of an approved IACUC protocol.

The initial data presented in this article were collected on rats being anesthetized for other animal studies. Rats were chosen over mice because of the greater volumes of isoflurane utilized and WAG produced. Confirmatory samples were later conducted on mice and a sham rat model.

Oxygen and isoflurane concentrations were initially evaluated at several locations including the rodent abdomen (incision site), nose cone, general area, and personal breathing zone. Samples collected at the incision site were within 5 cm from the rodent abdomen. Samples collected at the nose cone were approximately 5 cm from the rodent nose. General area samples were collected in an approximate radius of 1 meter from the surgery table at a height of 1.5 meters from the floor. Personal breathing zone samples were conducted with the sampler placed on the employee's lapel approximately 30 cm from his or her mouth and nose.

Oxygen concentrations were measured with a multi-gas air monitor (RKI Eagle, Union City, Calif.). The meter was used to measure concentrations of oxygen in units of percent. Isoflurane sampling was conducted with a miniature infrared ambient air analyzer (MIRAN SaphiRe XL, Thermo Fisher Scientific, Waltham, Mass.).⁽¹²⁾ This technology has been used previously for anesthetic gas monitoring.^(13,14) The infrared gas analyzer was a direct reading instrument, factory calibrated and programmed by the operator to measure isoflurane.⁽¹⁵⁾ The instrument's limit of detection ranged from 0.04–0.7 ppm with an accuracy of $\pm 10\%$, dependent on the pre-set path length. In this study 100 ppm was the high range limit as specified by the manufacturer.

Eight-hour, TWA personal air samples for isoflurane were captured with passive organic vapor monitors (SKC, Inc., Eighty Four, Penn., 15330). The monitors were sent to an American Industrial Hygiene Association-accredited laboratory (Datachem Laboratories, Inc., Cincinnati, Ohio) and analysis was performed on a Hewlett Packard 5890 gas chromatograph equipped with a flame ionization detector and a DB-624 capillary column.

RESULTS

Findings before Ventilation Controls

Initial findings in the laboratory animal surgery suite were elevated locally for oxygen and throughout the suite for isoflurane. Elevated oxygen levels were found at the rodent nose cone and incision site. Oxygen concentrations in the room were not affected by the localized increases and remained at 20.9%. Isoflurane levels were elevated locally at the nose cone, incision site, and within 30 cm of the induction chamber equipped with a passive charcoal filter. As noted in Table I, the isoflurane levels rose above the REL throughout the room during the surgical procedure. Concentrations from 51 ppm to over 100 ppm of isoflurane were measured at the nose cone and incision site. Variability at the nose cone can be explained by positioning and movement of the rodent during the procedure.

A significant finding was the contribution of isoflurane emissions to the room by the anesthetic induction process. In this process, the rodent was placed in the chamber to be initially anesthetized prior to placement on the rodent nose cone anesthesia system. The process of opening the chamber and moving the rodent in and out of the chamber emitted a significant amount of isoflurane into the room. A previous study found an excess of two times the excursion level for halothane (50 ppm) in the employee's breathing zone during the induction process.^(6,16)

The elevated oxygen findings were a flammability hazard and needed to be lowered to normal atmospheric concentrations (~20.9%). The recommended concentrations for isoflurane exposures to employees is less than 2 ppm as discussed in the Introduction section of this article.⁽²⁾ Measurements of isoflurane exceeded 25 times the REL for area excursions and

TABLE I. Air Sampling Summary Before Controls—During an Operation on a Live Rat

Sample Description	Analyte	Concentration
Nose cone area	O ₂	21.3–40% (n = 6, TWA = 5 sec) (μ = 23.8)
	Isoflurane	51– >100 ppm (n = 11, TWA = 10 sec) (μ = 89.8)
Incision site	O ₂	20.9– 33% (n = 6, TWA = 5 sec) (μ = 23.8)
	Isoflurane	>100 ppm (n = 6, TWA = 10 sec) (μ = >100)
Above induction chamber	Isoflurane	1.5–153 ppm (n = 10, TWA = 10 sec) (μ = 44.7)
Surgeon's breathing zone	Isoflurane	1.6–4.0 ppm (n = 4, TWA = 10 sec) (μ = 2.9)
General room during procedure	Isoflurane	1.2–8.0 ppm (n = 16, TWA = 10 sec) (μ = 3.4)

TABLE II. Employee Personal Breathing Zone 8-Hour TWA Air Sampling Summary During an Operation on a Live Rat

Sample Description	Analyte	Before Exhaust	After Exhaust
Surgeon	Isoflurane	5.3, 9.9 ppm n = 2	<0.1 ppm n = 2
Assisting Technicians	Isoflurane	1.9, 1.9 ppm n = 2	<0.1 ppm n = 2

almost 5 times the REL for personal 8-hour TWA samples as noted in Table II. Actual employee exposure varied from approximately 3–9 hours.

Corrective Actions

Various engineering controls such as chemical fume hoods, Class II type B2 biosafety cabinets, and numerous local exhaust ventilation options are available to control WAG.^(11,17) This facility, as with many laboratory animal operating rooms, has space and logistics that do not easily allow for ventilation controls such as fume hoods and biosafety cabinets.^(14,17–19) In this study, we focused on a control strategy that would be practical and cost-effective in controlling WAG during rodent surgical procedures.

To control emissions of both oxygen and isoflurane from the vaporizer, we designed a local exhaust nose cylinder, vented to the house vacuum system. Other researchers have designed modified nose cones with passive controls such as a latex shroud sometimes referred to as a diaphragm or balloon through which the rodent nose protrudes.⁽²⁰⁾ It is theorized that the shroud will prevent waste anesthetic gas from leaking past the rodents head as readily. In the authors' experience it is difficult to create a leak-proof seal on a rodent's nose and mouth even with a shroud. Another problem associated with the diaphragm-type nose cone is that it is typically associated with a passive control such as a charcoal canister relying on back pressure for scrubbing capabilities. We found the backpressure-type system to be inefficient, especially when used on an induction chamber. For these reasons we looked for active controls.

The clear plastic cylinder we designed was sized for mice 1–1/8 in inside diameter (I.D.) or 1–5/8 in I.D. for rats with a 1/4 in outside diameter (O.D.) side port for the vacuum line connection, as displayed in Figure 1. This active control was designed to evacuate the WAG without creating excessive negative pressure that would require increased gas flow to maintain anesthesia. The cylinder was sealed with tape at the vaporizer connection to reduce airflow at that site. The rodent face/head and the anesthetic delivery nose cone were enclosed within the vented gas scavenging cylinder, but there was no restriction of room airflow past the head of the rodent into the cylinder as shown in Figure 2.

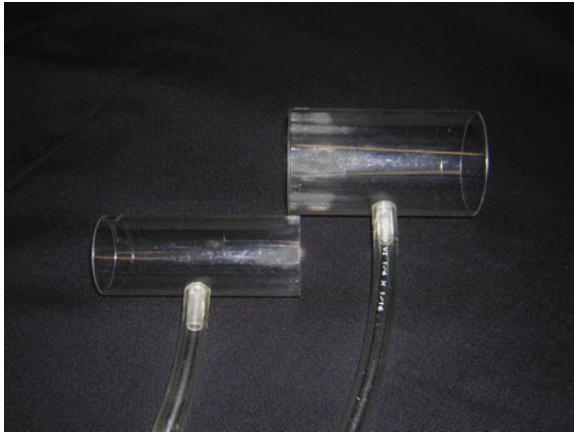


FIGURE 1. The mouse nose cylinder exhaust device pictured to the left. The rat nose cylinder exhaust device pictured to the right. (color figure available online)



FIGURE 2. Rat with nose cylinder exhaust in use. Note that tape is used to temporarily seal the cylinder to the gas delivery apparatus. (color figure available online)

Exhaust from the nose cylinder was measured with a precision rotameter (Dwyer Instruments, Inc., Michigan City, Ind., RMA-22-SSV). A minimum vacuum flow of 17 liters per minute provided adequate evacuation of waste oxygen and isoflurane when the rat's head was entirely contained in the cylinder as presented in Table II. An independent study using a similar scavenging system, found exhaust rates of 12–15 liters per minute to be acceptable.⁽¹¹⁾

Along with these engineering controls, it is important that excessive quantities of the anesthetic gas and oxygen are not administered. The volume of anesthetic vapor required by

laboratory rodents is minimal relative to the volume capability of most precision vaporizers, and a volume of approximately 500 ml/minute is more than adequate.⁽²⁰⁾ This was the gas flow rate used during cylinder testing.

There are several options to control emissions of isoflurane from the induction chamber including:

- an induction chamber contained and used in a chemical fume hood or externally exhausted biosafety cabinet
- an externally exhausted induction chamber
- a specialty local exhaust ventilation system such as slot vents or downdraft table.

TABLE III. Surgical Procedure Mean Isoflurane Concentrations

	Before Scavenging (ppm)		After Scavenging (ppm)		Reduction (percent)	
	In Vivo	In Vitro	In Vivo	In Vitro	In Vivo	In Vitro
Nose cone (TWA = 10 sec)	89.5 (n = 11)	15.55 (n = 6)	2.7 (n = 16)	0.15 (n = 6)	97	99
Incision site (TWA = 10 sec)	> 100 (n = 6)	8.28 (n = 6)	1.5 (n = 10)	0.03 (n = 6)	98	98
Surgeon breathing zone (TWA = 10 sec)	2.9 (n = 4)	0.24 (n = 5)	1.3 (n = 10)	0.13 (n = 6)	53	46
Room (TWA = 10 sec)	3.4 (n = 16)	0.18 (n = 5)	1.3 (n = 10)	0.12 (n = 6)	60	33

TABLE IV. Surgical Procedure Mean Oxygen Concentrations

	Before Scavenging (ppm)		After Scavenging (ppm)		Reduction to background (percent vol/vol)	
	In Vivo	In Vitro	In Vivo	In Vitro	In Vivo	In Vitro
Nose cone (TWA = 10 sec)	27.1 (n = 6)	21.6 (n = 6)	20.9 (n = 16)	20.9 (n = 6)	100	100
Incision site (TWA = 10 sec)	23.8 (n = 6)	20.9 (n = 6)	20.9 (n = 10)	20.9 (n = 6)	100	100

TABLE V. Induction Procedure Mean Isoflurane Concentrations (TWA = 10 sec)

	Before Scavenging (ppm)		After Scavenging (ppm)		Reduction (percent)	
	In Vivo	In Vitro	In Vivo	In Vitro	In Vivo	In Vitro
Above chamber	44.7 (n = 10)	83.8 (n = 4)	2.3 (n = 10)	1.4 (n = 4)	95	98
Room	3.4 (n = 16)	1.1 (n = 4)	1.3 (n = 10)	0.2 (n = 4)	60	84
Surgeon breathing zone	2.9 (n = 4)	10.2 (n = 4)	1.3 (n = 10)	1.4 (n = 4)	53	87

Given the space constraints of a veterinary surgical suite, we researched and recommended the use of an externally exhausted induction chamber. In the unit we chose, the animal is placed into the chamber via a sliding lid (Vented 2-liter Induction Chamber #942102, VetEquip, Inc., Pleasanton, Calif.).⁽²¹⁾ With the lid in the closed position, isoflurane is introduced into the chamber while a constant vacuum is applied above the chamber to capture excess gas as it leaves the induction area. When the lid is opened, the vacuum creates a negative pressure above the chamber, allowing safe removal of the animal and preventing excess vapor from emitting into the room (Figure 3). The chamber is vented similarly to a chemical fume hood at a rate of approximately 11.5 cubic feet per minute. It is important that the WAG is exhausted outside the building away from air intakes.⁽²²⁾

The authors recommended active controls because their testing, as displayed in Tables I and II, showed that both unabated surgical anesthesia and induction chambers equipped with passive charcoal filters did not control WAG to acceptable levels. When rodents are placed in the nose cone of a passive charcoal canister system, the nose does not make a tight seal, allowing WAG leakage. Also, when the lid on a passive charcoal canister induction system is opened to place the

animal into or take it out of the chamber, there is inadequate pressure to push the gas into the canister and the WAG escapes through the lid opening into the room.

Findings after Ventilation Controls

To test the effectiveness of the refined scavenging systems a sham rat procedure was designed. The surgical and induction procedures were duplicated using a commercially purchased sham rat. During the in vitro induction procedure, for comparison the anesthetic was administered through a passively vented induction chamber as well as the actively vented chamber. During the mock surgical procedure, the sham rat was administered anesthetic gases (EZ-Anesthesia, Euthanex Corporation, Palmer, Penn.) through the nose cone system, including the scavenging cylinder (Figure 4).

After implementation of the rodent nose cylinder and exhausted induction chamber controls, isoflurane concentrations in the laboratory animal surgery suite were significantly lower and below the REL, and oxygen levels were at a background of 20.9% at the rodent nose cone and incision site.

The in vivo results were similar to the in vitro reductions using the actively vented systems as noted in Tables III–V.



FIGURE 3. Exhausted induction chamber. The lower port supplies anesthetic gases. The upper port is vented to a drop pipe from an exhaust duct. To remove the anesthetized rodent from the charged chamber, the sliding lid opens under the hood flange, scavenging the gases. (color figure available online)



FIGURE 4. Sham rat with a canvas covering used in evaluations of the nose cylinder exhaust device for in vitro testing. (color figure available online)

TABLE VI. Induction Chamber Scavenging Comparison for Significance Using 95% Overlapping Means to Confidence Intervals

Location	Isoflurane Concentration Mean (ppm)	Significance Level	Standard Deviation	95% Confidence Interval
Above induction chamber without scavenging (n = 4)	83.8	0.05	68.9	16.5–152.7
Above induction chamber with scavenging (n = 4)	1.4	0.05	1.1	0.3–2.5
Breathing zone with chamber without scavenging (n = 4)	10.2	0.05	6.6	3.8–16.6
Breathing zone with chamber with scavenging (n = 4)	1.4	0.05	1.2	0.2–2.6
Room with chamber without scavenging (n = 4)	1.1	0.05	0.66	0.4–1.7
Room with chamber with scavenging (n = 4)	0.2	0.05	0.1	0.1–0.3

In vivo isoflurane reductions for the induction chamber emissions, the operating room, and the surgeon's breathing zone were 95%, 60%, and 53%, respectively. The same measurements for the in vitro (sham) procedure were 98%, 84%, and 87%, respectively. The increased reductions for the in vitro study are attributed to the controlled conditions of the study.

The induction process yields the greatest effect on controlling isoflurane exposures at the surgeon's breathing zone and in the overall room concentrations. Thus, capturing isoflurane emissions during the induction process becomes the most important isoflurane control during rodent surgery. Table VI shows that there is a significant decrease in isoflurane concentrations at the induction chamber, in the surgeon's breathing zone, and in the operating room when using the exhausted induction chamber. Significance was determined using the conservative overlapping confidence interval method of hypothesis testing at a 95% confidence interval.⁽²³⁾

DISCUSSION AND CONCLUSION

The laboratory animal surgeries using anesthetic gases in this study showed excessive concentrations of isoflurane throughout the surgical suite. Oxygen levels, measured 21.3–40% in the surgical field, posed a flammability hazard due to electrocautery and other ignition sources. Isoflurane concentrations exceeded 50 times the REL for area excursions and almost five times the REL for personal TWA samples when passive or no controls were used for capturing the waste gases.

The implementation of an exhausted rodent nose cylinder and exhausted induction chamber significantly lowered the isoflurane and oxygen levels to acceptable concentrations. Oxygen levels were no longer at an enriched concentration and isoflurane concentrations were below recommended exposure limits.

In vitro (sham) testing compared favorably to in vivo measurements, validating the reduction capability of active ventilation during rodent anesthetic administration.

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REFERENCES

1. **Thomann, W.R.:** Chemical safety in animal care, use, and research. *ILAR* 44:13–19 (2003).
2. **National Institute for Occupational Safety and Health (NIOSH):** *Criteria for a Recommended Standard: Occupational Exposure to Waste Anesthetic Gases and Vapors*, Publication no. 77–140. Cincinnati, OH: NIOSH, 1977.
3. **Berry, A.J.:** Recommended exposure limits for desflurane and isoflurane. *Anesth. Analg.* 88:1424–1425 (1999).
4. **Sessler, D.I., and J.M. Badgwell:** Exposure of postoperative nurses to exhaled anesthetic gases. *Anesth. Analg.* 87:1083–1088 (1998).
5. "Request for Information on Waste Halogenated Anesthetic Agents: Isoflurane, Desflurane, and Sevoflurane," *Federal Register* 71:8859–8860. 2006.
6. "Airborne Contaminants: Dusts, Fumes, Mists, Vapors, and Gases," *California Code of Regulations. Title 8, Section 5155, Subchapter 7, Group 16, Article 107.*
7. **Klassen, C.D.:** *Casarett & Doull's Toxicology: The Basic Science of Poisons*. New York: McGraw Hill Medical Publishing Division. 6th ed. 2001.
8. **Smith, J.C., and B. Bolon:** Atmospheric waste isoflurane concentrations using conventional equipment and rat anesthesia protocols. *Contemp. Top Lab. Anim. Sci.* 41(2):10–17 (2002).
9. **Danulescu, E., M. Ghitescu, D. Constantinescu, et al.:** Potential health effects within occupational exposure to anesthetics of health care workers. *J. Prev. Med.* 14:17–25 (2006).
10. **Klein, R.C.:** Fire safety recommendations for administration of isoflurane anesthesia in oxygen. *Lab. Anim.* 37:223–224 (2008).
11. **Canright M., and B. Kollmeyer:** Thinking Outside the Nosecone: A Case Study Discussion a Novel Solution to Control Waste Anesthetic Gas Exposure in a Vivarium. *American Industrial Hygiene Conference and Exposition Proceedings, CS-116-02*. Cincinnati, OH: ACGIH® (2012)
12. **Thermo Electron Corporation:** MIRAN SapphIRe Analyzer. Ventilation Studies Using Tracer Gas, Publication no. LAD 001-037. Franklin, MA: Thermo Electron Corporation.
13. **Burkhart, J.E., and T.J. Stobbe:** Real-time measurement and control of waste anesthetic gases during veterinary surgeries. *Am. Ind. Hyg. Assoc. J.* 51:640–645 (1990).

14. **Smith, J.C., and B. Bolon:** Isoflurane leakage from non-rebreathing rodent anaesthesia circuits: Comparison of emissions from conventional and modified ports. *Lab. Anim.* 40:200–209 (2006).
15. **Rasmussen H., and S. Thorud:** Using a refrigerant leak detector to monitor waste gases from halogenated anesthetics. *J. Amer. Assoc. Lab. Anim. Sci.* 46:64–68 (2007).
16. **Korczynski, R.E.:** Anesthetic gas exposure in veterinary clinics. *Appl. Occup. Environ. Hyg.* 14:384–390 (1999).
17. **Cooper, D.M., D.C. Errede, and A. Streifel:** Assessment of occupational exposure to isoflurane administered in an anesthetic chamber within a horizontal laminar flow clean bench. *Contemp. Top. Lab. Anim. Sci.* 37:64–67 (1998).
18. **McGregor, D.G.:** Occupational exposure to trace concentrations of waste anesthetic gases. *Mayo Clin. Proc.* 75:273–277 (2000).
19. **Taylor D.K., and D.M. Mook:** Isoflurane waste anesthetic gas concentrations associated with the open-drop method. *J. Amer. Assoc. Lab. Anim. Sci.* 48:61–64 (2009).
20. **Smith, J.C.:** Waste Anesthetic Gas Safety. In *Anesthesia and Analgesia in Laboratory Animals*, R.E. Fish, M.J. Brown, P.J. Danneman, and A.Z. Karas. Amsterdam: Elsevier, 2008. 2nd ed., pp. 183–193.
21. **TSS/Vet Equipment:** Vented 2-Liter Induction Chamber. Project NC040NPNA-04, Report 156. Pleasanton, CA, 2004.
22. **American Conference of Governmental Industrial Hygienists (ACGIH®):** *Industrial Ventilation: A Manual of Recommended Practice*, 25th ed. Cincinnati, OH: ACGIH Worldwide Signature Publications, 2004.
23. **Payton, M.E., M.H. Greenstone, and N. Schenker:** Overlapping confidence intervals or standard error intervals: What do they mean in terms of statistical significance? *J. Insect. Sci.* 3:34 (2003).